Deck Pro

AI Deck Condition Analysis System

Problem Statement, Data Requirements and Quality Specification

Document	Problem statement (version 2, expanded)
Domain	Machine learning, computer vision, dataset preparation
Audience	ML engineer, full-stack developer, product owner, restoration expert
Core task	Image-quality gate, then condition classification, then structured report
Status	Ready for pilot planning; all numeric thresholds are starting values to be tuned on real data

In one line: a user uploads deck photos; the system first checks whether the photos are good enough to judge, and only then predicts the visible condition with a calibrated confidence. Poor, irrelevant or uncertain images are never forced into a label; they are sent back for a better photo or for human review.

1. Purpose and the problem in plain words

Wooden decks wear out from sun, rain, moisture, foot traffic and the slow failure of stain or sealer. Before a restoration job is quoted, someone has to look at photos and judge how bad the deck looks. Today that judgement is manual, slow, and differs from person to person. Two reviewers can look at the same photo and call it 'mild' and 'moderate'.

Deck Pro needs a dependable first-level visual assessment tool. A customer or contractor uploads photos; the system tells them, in a consistent and explainable way, what visible condition the deck appears to be in and what to inspect next. It supports the existing estimator. It does not replace a professional inspection and it does not set prices.

> The quality bar is the key point of this version of the document. A model that answers confidently on every image is cheap to build and dangerous to trust. We want a model that is right when it speaks and honest when it cannot tell. That requires three things working together: good data, an image-quality gate in front of the model, and an uncertainty gate behind it.

1.1 Objectives

- Classify the overall visible condition of a wooden deck into five levels, using a trained image model rather than a per-image LLM or vision API.
- Detect which specific visible conditions are present (fading, peeling, cracking, dark staining, and so on).
- Verify that each photo is clear and informative enough before any prediction is made, and tell the user exactly how to fix it when it is not.
- Return a predictable, structured result with calibrated confidence, limitations and a manual-review flag.
- Map predictions to expert-approved recommendation text, never to model-written advice or prices.
- Keep every result traceable to a model version, an image and an estimate.

1.2 Out of scope

- Structural safety judgements (rot depth, load capacity, fastener or joist condition, railing safety). These cannot be determined from a surface photo.
- Pricing, area measurement and quantity take-off. The approved estimator formulas stay in charge.
- Diagnosing the cause of dark patches (moisture, mould, tannin, dirt). The system reports what is visible, not why.
- Free-form conversational explanations. An LLM may be considered later, but it is not part of the primary workflow.

2. End-to-end workflow

Step	Stage	What happens	If it fails
1	Upload	User submits one or more photos (JPEG, PNG, HEIC converted server-side). File size, type and corruption checks.	Reject with a clear message
2	Image quality gate	Automated checks for resolution, sharpness, exposure, glare, deck coverage and viewing angle (section 3).	Ask for a better photo with specific advice
3	Domain check	Confirm the image actually shows a wooden deck surface and not a fence, interior floor, stone patio or person.	Mark as unsupported, request a new photo

Step	Stage	What happens	If it fails
4	Classification model	Predict overall level 1 to 5 and multi-label visible conditions, each with a calibrated confidence.	-
5	Prediction gate	Check confidence, margin between top two classes, and out-of-distribution score.	Return 'Unable to assess' and route to manual review
6	Multi-photo aggregation	Combine results across photos of the same deck conservatively (section 6).	Flag disagreement for review
7	Condition report	Display category, visible evidence, limitations and approved next steps.	-
8	Estimator hand-off	Estimator uses verified measurements and approved pricing rules. The image result is advisory input only.	-

3. Image clarity and minimum quality requirements

A deck-condition label is only as reliable as the photo behind it. Fading, early peeling and fine cracks are small, low-contrast features. If the image is blurry, dark, tiny or taken from too far away, even a perfect model is guessing. For that reason the system must measure image quality first and refuse to classify below a minimum standard. This gate applies to training data as well as to live uploads, so the model never learns from images a human could not judge either.

3.1 Automated quality checks

Check	What it protects against	How it is measured (starting approach)	Starting threshold (tune on data)
Resolution	Detail too small to see peeling or cracks	Pixel width and height of the original, before any resizing	Shorter side at least 1000 px; reject below 640 px. Never judge a photo that was upscaled.
Sharpness / blur	Motion blur, missed focus	Variance of Laplacian on the deck region; optional FFT high-frequency energy; compare to a per-resolution baseline	Reject below a threshold calibrated so that human reviewers also call those images 'unreadable'
Exposure	Too dark or washed out	Histogram clipping: share of near-black and near-white pixels; mean luminance	Reject if more than about 25 percent of deck pixels are clipped
Glare and harsh shadow	Reflections or dappled shade hiding surface texture	Detect large saturated patches and high local contrast stripes	Warn, or reject if over a set share of the deck area
Deck coverage	Photo mostly sky, house, furniture or lawn	Segmentation or detector gives the fraction of the frame that is deck boards	Deck boards fill at least about 30 percent of the frame for full views; at least 50 percent for close-ups
Viewing angle	Extreme grazing angle where boards are unreadable	Perspective estimate from board-line orientation	Warn when near-horizontal; prefer 30 to 90 degrees down onto the surface
Occlusion	Furniture, rugs, snow, leaves, pots covering the surface	Detect non-deck objects over the deck region	Warn or reject if occlusion is large
Compression and artifacts	Heavily compressed or s creenshot-of-a-screensh ot images	JPEG quality estimate, blockiness, EXIF origin	Warn when quality is very low
Duplicates	Same photo uploaded repeatedly	Perceptual hash	Merge duplicates

These thresholds are starting hypotheses. The correct way to set them is empirical: have restoration reviewers mark a few hundred images as 'readable' or 'not readable', then pick the thresholds that best reproduce those human judgements. A learned image-quality classifier can later replace hand-written rules.

3.2 Two-level quality outcome

- **Pass:** good enough to classify; continue to the model.
- **Pass with warning:** classifiable but with reduced reliability; the report states the limitation and may lower the confidence shown.

- **Fail:** not classified. The user sees a specific reason (for example 'Image is too blurry. Hold the phone steady and tap the screen to focus') and a request to retake it.

3.3 Capture guidance shown to users

Most bad images come from bad capture, so the interface should coach the user before and after upload.

- Take one wide photo of the whole deck and several close-ups of the worst areas.
- Stand or hold the camera so boards fill most of the frame, roughly 1 to 1.5 m above the surface for close-ups.
- Photograph in even daylight, early morning or overcast is ideal. Avoid direct glare and deep shadow.
- Tap to focus on the boards and hold still. Do not use digital zoom, filters or beauty modes.
- Move furniture, mats and debris out of the shot where possible.
- Upload the original photo, not a screenshot or a messaging-app copy that has been compressed.

3.4 Why this matters for training

- Apply the same quality gate to every candidate training image and record the quality status in the dataset.
- Keep a small, clearly labelled set of poor-quality and irrelevant images so the model and the gate learn to recognise them.
- Evaluate model accuracy separately on 'good', 'borderline' and 'poor' quality subsets. A strong overall score that hides weak performance on borderline photos is not acceptable.
- Data augmentation (blur, brightness, noise, compression) should simulate realistic phone conditions but must not be used to manufacture labels for images a human cannot judge.

4. Categories to be recognised

The model only learns what it is taught, so labels must be specific, consistent and reviewed by someone with deck restoration knowledge. Three outputs are used instead of one combined label.

4.1 Overall visible condition level (single label)

Level	Label	Meaning (visual only)
1	Good condition	Surface looks relatively intact with minimal visible deterioration.
2	Mild wear	Minor fading, light discoloration or limited coating wear.
3	Moderate deterioration	Noticeable fading, coating wear or localised peeling.
4	Severe deterioration	Extensive peeling, substantial coating loss or widespread surface deterioration.
5	Significant visible wood damage	Prominent cracks, splintering or other apparent wood damage that needs closer inspection.

These are visual condition categories, not engineering safety ratings. A good-looking surface says nothing about whether the structure is sound.

4.2 Visible condition types (multi-label)

- Fading or discoloration
- Peeling or flaking coating
- Cracks or splintering
- Dark staining or suspected moisture-related discoloration (cause not confirmed)
- No major visible deterioration

4.3 Image validity (single label)

- Valid deck image, good quality
- Valid deck image, borderline quality (usable with warning)
- Poor quality (blurry, dark, too far, occluded)
- Unsupported or irrelevant (not a wooden deck, wrong material, or outside trained scope)

4.4 Handling ambiguity between levels

Adjacent levels (for example 2 versus 3) are inherently fuzzy. Treat the levels as **ordinal**: predicting level 4 when the truth is 3 is a smaller error than predicting level 1. Use an ordinal-aware loss or metric (such as quadratic weighted kappa and mean absolute level error) in addition to per-class precision and recall. Write a labeling guide with reference photos for every level and every boundary case, and revise it whenever reviewers disagree.

5. Dataset: collection, labeling and splitting

5.1 Sources, in order of value

Source	Value	Notes
Deck restoration contractors	Highest	Before and after photos, close-ups of defects and condition labels confirmed by professionals. Obtain written permission to use the images for model development.
Deck Pro customer uploads	High over time	With explicit consent, keep selected images and verified labels to build a proprietary, representative dataset.
Public datasets (Kaggle, Roboflow Universe, Hugging Face, academic collections)	Medium to low	Often indoor timber or industrial wood defects. Verify license, labels, image quality and domain match before use. Discovery sources only; a ready-made outdoor deck dataset may not exist.
Synthetic or generated images	Supplement only	May help rare classes but must never replace real data and must never appear in the test set.

5.2 How many images

Stage	Planning target
Pipeline prototype	100 to 200 images per category
More useful training set	300 to 500 images per category (about 1,500 to 2,500 for five levels)
Stronger real-world coverage	1,000+ images per category where feasible

These are planning figures, not guarantees. Required volume depends on how separable the categories are, variety in light, angle, wood type and stain colour, and above all label quality. Training successfully does not mean the model is production ready.

5.3 Required variety in the data

- Wood types and species look: pressure-treated pine, cedar, redwood, hardwoods, and, if in scope, composite (decide explicitly).
- Stain and coating colours from clear to solid, new and aged.
- Lighting: sun, shade, overcast, dawn and dusk, wet and dry boards.
- Cameras: low-end and high-end phones, different lenses and zoom levels.
- Distances and angles: full views, mid views, close-ups.
- Regions and climates, so that weathering patterns are not learned from one place only.
- Hard negatives: a pristine deck that looks good but is incomplete, freshly stained wet wood, shadows that mimic dark stains, dirt versus damage.
- Many different decks. Many near-identical photos of one deck do not count as independent examples.

5.4 Annotation record for every image

- Image ID, source and permission status
- Overall condition level (1 to 5)
- Visible condition labels (zero or more)
- Image quality status and the measured quality scores

- Deck or project ID (for grouped splitting)
- Reviewer ID, expert-verified flag and optional uncertainty note
- Capture metadata where available: device, date, lighting note

5.5 Labeling quality control

- Write the labeling guide first, with reference images for each level and boundary.
- Have two or more reviewers label an overlapping sample independently. Measure agreement (Cohen's or Fleiss' kappa, plus ordinal agreement). If agreement is low, fix the guide or merge categories before labeling more.
- Send disagreements and low-agreement images to a restoration expert for adjudication.
- Run confident-learning or loss-based audits after the first model to find likely mislabeled images, then have an expert re-check them.
- Version the dataset and the labeling guide so every model can be tied to the exact data it was trained on.

5.6 Splitting without leakage

Use roughly 70 percent training, 15 percent validation and 15 percent test, but split **by deck or project**, never by image. All photos of one deck must stay in one split. Keep a separate final test set that is touched only once, plus, when possible, a **site-held-out** set from contractors or regions never seen in training. Stratify so each level and each quality tier appears in every split.

6. Model approach

6.1 Recommended design

- Transfer learning from a pretrained image model, not a CNN trained from scratch.
- Candidates: MobileNetV3 (small, cheap to serve) and EfficientNet (benchmark both on your data; do not assume which wins).
- Two heads on a shared backbone: a 5-level ordinal or classification head and a multi-label head for visible conditions.
- Train at a resolution high enough to see fine detail. Consider tiling large images into crops for fine features, then aggregating.
- Handle class imbalance with weighted loss or balanced sampling; severe cases are usually rarer.
- Use realistic augmentation (rotation, crop, colour jitter, blur, noise, compression), validated so it does not erase the very features that define a class.
- Fine-tune in stages: train the new heads first, then unfreeze upper layers with a small learning rate.
- Keep a simple baseline (for example a frozen backbone with logistic regression) so the benefit of fine-tuning is measured, not assumed.

6.2 Confidence, calibration and rejection

A softmax score of 0.90 does not mean a 90 percent chance of being right. Calibrate confidence on held-out data using temperature scaling or similar, and report calibration error. Then build a rejection path with three signals: low top-class confidence, small margin between the top two classes, and a high out-of-distribution score (for example from embedding distance or an energy score). When any signal trips, the system returns 'Unable to assess' and requests a better photo or manual review. Choose the thresholds from a risk-coverage curve agreed with the product owner, for example 'at least X percent accuracy on the images we do answer, covering at least Y percent of uploads'.

6.3 Multiple photos of one deck

- Run the quality gate and model on each photo independently.
- Aggregate conservatively. For the overall level, weight by image quality and confidence, and surface the worst credible area rather than averaging it away.
- If photos disagree by two or more levels, flag for manual review.
- Show the user which photo produced which finding.

6.4 Explainability

Provide visual evidence such as Grad-CAM or similar heatmaps for review and debugging. Treat them as an inspection aid, not proof of correctness. The report text itself comes from approved templates keyed to the predicted labels.

7. Evaluation and acceptance criteria

Do not set a single arbitrary accuracy target in advance. Agree targets with the product owner and restoration experts after the first pilot, with special attention to severe conditions and to uncertain cases.

What to measure	Why
Per-class precision, recall and F1; macro-F1; confusion matrix	Overall accuracy can hide poor performance on severe classes.
Ordinal metrics: mean absolute level error, quadratic weighted kappa, share within one level	Being off by one level is not the same as being off by three.
Multi-label precision, recall and F1 per visible condition	Each feature can fail independently.
Calibration: expected calibration error, reliability diagram	Confidence shown to users must be honest.
Risk-coverage curve and rate of 'Unable to assess'	Quantifies the trade-off between answering and abstaining.
Accuracy by quality tier (good, borderline, poor)	Proves the quality gate is doing its job.
Accuracy by wood type, stain colour, lighting, device and region	Reveals hidden weak spots and bias in the data.
Quality-gate agreement with human reviewers (precision and recall of 'unreadable')	A gate that rejects good photos frustrates users; one that accepts bad photos lets errors through.
Human comparison: model versus two expert reviewers on the same unseen decks	A human-level benchmark gives the target a real meaning.
Robustness tests: rotated, cropped, compressed, noisy and relit versions	Checks stability to ordinary phone variation.
Latency and cost per image	Confirms the cost advantage over per-call vision APIs.

8. System architecture and API

- **React:** upload with live capture guidance, progress, result display and report history.
- **Node.js + Express:** authentication, roles, validation, rate limiting, and integration with the existing estimator.
- **Python + FastAPI ML service:** quality gate, preprocessing, model inference, calibration and rejection logic. The trained model is loaded once at start-up, never retrained per upload.
- **PyTorch or TensorFlow:** training and inference.
- **MongoDB:** reports, image references, model versions and estimate links. Images themselves live in secure object storage with access control, not inside MongoDB.
- The existing MERN application and the pricing engine remain unchanged and independent.

Illustrative response (values are examples, not measured performance):

```
{
  "image_id": "DECK-00042",
  "model_version": "deck-condition-v1",
  "status": "completed",
  "image_quality": {
    "status": "pass_with_warning",
    "resolution_ok": true, "sharpness_score": 0.78,
    "exposure_ok": true, "deck_coverage": 0.64,
    "warnings": ["Mild glare in upper right"]
  },
  "overall_condition": {
    "level": 3, "label": "Moderate deterioration",
    "confidence": 0.87, "calibrated": true
  },
  "visible_conditions": [
    {"label": "peeling_coating", "confidence": 0.91},
    {"label": "fading", "confidence": 0.84}
  ],
  "limitations": ["Structural safety cannot be determined from this image."],
  "requires_manual_review": false,
  "rejection_reason": null
}
```

The API must also define responses for rejected images (with a human-readable reason and retake advice), low-confidence results, partial failures in multi-photo uploads and service errors. Every response carries the model version and quality-gate version.

9. Report content and safety wording

- Show the level, label, confidence, visible features and the photos used.
- Show approved recommendation text mapped from the predicted labels, for example 'inspect coating adhesion and determine the surface preparation needed before recoating'.
- Always state limitations: this is a visual assessment, not a structural inspection, and a professional inspection is still required.
- Never prescribe sanding or replacement for every surface, never declare the wood unsafe, and never output a price from the image alone.
- When the result is 'Unable to assess', say why and what the user can do (retake photos, add close-ups, request an inspection).
- Wording must be reviewed by the restoration expert and, where relevant, legal or compliance advisers.

10. Privacy, consent and security

- Obtain clear consent before storing customer images for training, and let users opt out.
- Strip or restrict EXIF location data, and blur faces, house numbers and licence plates if present.
- Encrypt images in storage and in transit; apply role-based access and signed, expiring URLs.
- Define retention periods and a deletion process; honour deletion requests across storage, datasets and backups where required by applicable law.
- Validate uploads against malicious files; limit size and rate.
- Record licenses for every external dataset and keep a data lineage log.

11. Monitoring, human review and continuous improvement

- Log every prediction with model version, quality scores and outcome, without storing more personal data than needed.
- Provide a review queue for 'Unable to assess', low-confidence and disagreement cases. A restoration expert's decision becomes a verified label.
- Track drift: change in quality-score distribution, confidence distribution, rejection rate and class mix per month.
- Compare predictions with contractor findings on completed jobs, with permission, to measure real-world accuracy.
- Retrain on a schedule or when drift appears, evaluate on the frozen test set and a fresh site-held-out set, and release with a model card that records data, metrics and known limitations.
- Use shadow mode and then a limited pilot before full release; keep the ability to roll back to a previous model version quickly.

12. Risks and mitigations

Risk	Mitigation
Too little or unrepresentative data	Start with a pilot, prioritise contractor data, track coverage across wood types, lighting and regions.

Risk	Mitigation
Inconsistent labels between levels	Labeling guide with reference images, double labeling, expert adjudication, ordinal metrics.
Data leakage inflating results	Split by deck or project; keep an untouched final test set and a site-held-out set.
Overconfident wrong predictions	Calibration, margin and out-of-distribution rejection, mandatory manual-review path.
Poor user photos	Quality gate, live capture guidance, specific retake messages, multi-photo support.
Model mistaken for a safety inspection	Explicit limitations, out-of-scope statement, wording review.
Bias towards certain wood or stain types	Stratified evaluation, targeted data collection, subgroup reporting.
Drift as cameras and decks change	Monitoring, periodic retraining, versioned releases.
Legal or consent issues with images	Written permissions, consent flow, license tracking, retention policy.

13. Delivery plan

Phase	Work	Output
0. Definition	Finalise labels, labeling guide, quality criteria and success metrics with a restoration expert and the product owner.	Signed-off specification
1. Data pilot	Collect a few hundred real decks, apply the quality gate, double-label a sample, measure agreement.	Pilot dataset v0, agreement report
2. Baseline	Train the baseline and a first transfer-learning model; evaluate on a deck-grouped split.	Baseline report, error analysis
3. Quality gate and rejection	Tune the image-quality thresholds against human judgements; add calibration and out-of-distribution rejection.	Gate and calibration report
4. Integration	FastAPI service, Node.js integration, report templates, storage and review queue.	Working end-to-end prototype
5. Controlled pilot	Run with a few contractors in shadow mode, then live with manual review.	Pilot results against contractor findings
6. Scale and monitor	Expand data, add feature labels, retrain, monitor drift.	Model v1.x with model card

14. Definition of done for the first controlled pilot

- Users can upload supported formats securely, and invalid or corrupt files are handled.
- The image-quality gate rejects or warns on blurry, dark, tiny, glared, occluded, irrelevant and unsupported images, and its behaviour is validated against human reviewers.
- The model predicts the agreed levels and visible conditions, with calibrated confidence.
- Low-confidence, ambiguous and out-of-distribution images return 'Unable to assess' and can be sent for manual review.
- Performance is documented separately for each class, each quality tier and key subgroups.
- The model has been tested on photographs of decks never seen in training, including a site-held-out set.
- The report shows reviewed recommendations and clear limitations, and contains no structural-safety claim or price.

- Every prediction is traceable to the image, model version, quality-gate version and estimate.
- Consent, retention and security controls are in place.
- The existing estimator and pricing rules continue to run independently.

15. Glossary

Term	Meaning
Transfer learning	Starting from a model already trained on a large image collection and adapting it to deck photos.
Calibration	Making confidence scores match how often predictions are actually correct.
Out-of-distribution (OOD)	An input unlike anything the model was trained on, where its answer cannot be trusted.
Ordinal labels	Labels with a natural order, where being one level off is a smaller error than being four off.
Data leakage	Near-duplicate or same-deck images appearing in both training and test data, making results look better than reality.
Laplacian variance	A simple sharpness measure: low values indicate a blurry image.
Risk-coverage curve	A plot showing accuracy against the share of images the system is willing to answer.
Model card	A short document recording a model's data, intended use, metrics and limitations.

Final recommendation: build Deck Pro's assessment as a dedicated, lightweight computer-vision service. Start with definitions, data and the image-quality gate, then train with transfer learning, calibrate, and prove the model on unseen real decks before connecting it to the estimator. A system that says 'please retake this photo' or 'a human should look at this' at the right moments will earn far more trust than one that always answers.